Question	Answer	Marks	Partial Marks
7(c)(i)	$[y =]\ -\frac{1}{4}x+\frac{23}{8}$ oe	**3**	**B2FT** for $[y =]\ -\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}x+c$ oe (c can be numeric or algebraic) OR **M2** for $\frac{y-2}{x-3.5}=-\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}$ oe OR **M1** for $-\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}$ soi **M1** for correct substitution of (3.5, 2) into $y = (\textit{their } m)x + c$ oe
7(c)(ii)	(–4.5, 4)	**2**	**B1** for each value or for $\begin{pmatrix}-8\\2\end{pmatrix}$ seen
8(a)(i)	$\frac{x-1}{x+2}$	**2**	**B1** for either numerator or denominator correct
8(a)(ii)(a)	$\frac{x}{x+3}\times\frac{x-1}{x+2}=\frac{7}{15}$	**B1**	**FT** *their* **(a)(i)** = $\frac{7}{15}$
	$15x(x – 1) = 7(x + 3)(x + 2)$	**M1**	Removes all algebraic fractions **FT** *their* equation if in comparable form
	$15x^2 – 15x = 7x^2 + 21x + 14x + 42$	**M1**	Correctly expands all brackets **FT** *their* equation if in comparable form
	$[8x^2 – 50x – 42 = 0]$ $4x^2 – 25x – 21 = 0$	**A1**	With no errors or omissions seen and one further stage seen after final M1
8(a)(ii)(b)	$(4x + 3)(x – 7)$ [= 0]	**M2**	**M1** for $4x(x – 7) + 3(x – 7)$ or $x(4x + 3) – 7(4x + 3)$ or for $(4x + a)(x + b)$ where either $ab = –21$ or $4b + a = –25$ If 0 scored, **SC1** for $4x + 3$ **and** $x – 7$ seen but not in factorised form
	7 and $-\frac{3}{4}$	**B1**	
8(a)(ii)(c)	7	**1**	**FT** *their* positive solution

Question	Answer	Marks	Partial Marks
8(b)	$\frac{1}{6}$ oe	**4**	**M3** for $\frac{5}{9}\times\frac{4}{8}\times\frac{3}{7}+\frac{4}{9}\times\frac{3}{8}\times\frac{2}{7}$ or **M2** for $\frac{5}{9}\times\frac{4}{8}\times\frac{3}{7}$ or $\frac{4}{9}\times\frac{3}{8}\times\frac{2}{7}$ or **M1** for $\frac{5}{9},\frac{4}{8},\frac{3}{7}$ seen or $\frac{4}{9},\frac{3}{8},\frac{2}{7}$ seen If 0 scored, **SC1** for $\frac{5^3+4^3}{729}$ oe
9(a)(i)	$\angle ACD = 46$ soi or $\angle CDE = 44$ soi	**B2**	**B1** for angle $ADC = 108$ or angle $DCB = 18$
	$\frac{58\sin 108}{\sin \textit{their} 46}$	**M2**	**M1** for $\frac{\sin 108}{x}=\frac{\sin \textit{their} 46}{58}$ oe
	76.68… nfww	**A1**	
9(a)(ii)	10.9 or 10.91 to 10.94	**3**	**B2** for [AB =] 68.9 or 68.91 to 68.94 or **M2** for a correct explicit statement for AB or BD or **M1** for $\frac{AB}{76.7}$ = cos26 oe
9(b)(i)	10.4 or 10.43 to 10.44	**4**	**M3** for $\sqrt{\frac{70}{\sin 40}}$ oe or **M2** for $x^2 \times \sin 40 = 70$ oe or **M1** for $\frac{1}{2}x \times 2x \times \sin 40 = 70$
9(b)(ii)	140	**1**	
10(a)(i)	3, –1	**2**	**B1** for each
10(a)(ii)	$23 - 4n$ oe final answer	**2**	**M1** for $k - 4n$ or $23 - jn$ ($j \neq 0$)
10(a)(iii)	22	**2**	**M1** for *their* **(a)(ii)** = –65
10(b)	23	**2**	**B1** for 37 or 60